Power equations

1 Solve the following equations:

a $x^5 = 3^5$ **b** $8^{-7} = x^{-7}$ **c** $x^3 = \left(\frac{8}{5}\right)^3$ **d** $x^{-7} = \frac{1}{6^7}$

e $3\left(x^{-9}\right) = \frac{3}{2^9}$ **f** $x^{\frac{1}{3}} = \sqrt[3]{6}$ **g** $\sqrt[3]{5} = x^{\frac{1}{3}}$ **h** $\frac{1}{2^5} = x^{-5}$

Exponential equations

2 Solve the following exponential equations:

a $4^x = 4^8$ **b** $3^x = 3^{\frac{2}{9}}$ **c** $3^x = 27$ **d** $7^x = 1$

e $8^x = \frac{1}{8^2}$ **f** $3^y = \frac{1}{27}$ **g** $7\left(4^x\right) = \frac{7}{4^3}$ **h** $5^x = \sqrt[3]{5}$

i $30^n = \sqrt[3]{30}$ **j** $10^x = 0.01$

3 Consider the following equations:

i Rewrite each side of the equation with a base of 2.

ii Solve for x.

a $8^x = 4$ **b** $16^x = \frac{1}{2}$ **c** $\frac{1}{1024} = 4^x$ **d** $\left(\sqrt{2}\right)^x = \sqrt[5]{32}$

4 Solve the following exponential equations:

a $\left(\sqrt{6}\right)^y = 36$ **b** $\left(\sqrt{2}\right)^k = 0.5$

c $9^y = 27$ **d** $3^{5x-10} = 1$

e $25^{x+1} = 125^{3x-4}$ **f** $\frac{1}{3^{x-3}} = \sqrt[3]{9}$

g $\left(\frac{1}{9}\right)^{x+5} = 81$ **h** $\left(\frac{1}{8}\right)^{x-3} = 16^{4x-3}$

i $\frac{25^y}{5^{4-y}} = \sqrt{125}$ **j** $8^{x+5} = \frac{1}{32\sqrt{2}}$

k $30 \times 2^{x-6} = 15$ **l** $2^x \times 2^{x+3} = 32$

m $3^x \times 9^{x-k} = 27$ **n** $a^{x-1} = a^4$

o $a^{x+1} = a^3\sqrt{a}$ **p** $3^{x^2-3x} = 81$

q $27\left(2^x\right) = 6^x$ **r** $3^x \times 3^{nx} = 81$

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Exponential equations and logarithms

5 Consider the following equations:

i Rearrange the equation into the form $x = \dfrac{\log A}{\log B}$.

ii Evaluate x to three decimal places.

a $13^x = 5$ **b** $5^x = \dfrac{1}{11}$ **c** $3^x = 2$ **d** $4^x = 6.4$

e $(0.4)^x = 5$ **f** $5^x + 4 = 3129$ **g** $2^{-x} = 6$ **h** $27^x + 4 = 19\,211$

6 For each of the following incorrect sets of work:

i Which step was incorrect? Explain your answer.

ii Rearrange the original equation into the form $a = \dfrac{\log A}{\log B}$.

iii Evaluate a to three decimal places.

a

$$\begin{aligned} 9^a &= 40 \\ \log 9^a &= \log 40 && (1) \\ a + \log 9 &= \log 40 && (2) \\ a &= \log 40 - \log 9 && (3) \\ &\approx 0.648 && (4) \end{aligned}$$

b

$$\begin{aligned} 2^a &= 89 \\ \log 2^a &= \log 89 && (1) \\ a \log 2 &= \log 89 && (2) \\ a &= \log_{89} 2 && (3) \\ &\approx 0.154 && (4) \end{aligned}$$

7 Find the x-coordinate of the point of intersection of the graphs of $y = 2^{5x}$ and $y = 4^{x-3}$.

Logarithmic equations

8 Solve for y in each of the following logarithmic equations:

a $\log_7 y = 5$ **b** $\log_{16} y = \dfrac{1}{2}$

c $\log_y 8 = 3$ **d** $2 \log_2 y + 3 \log_2 y = 15$

e $\log_6 y^2 = 4$ **f** $\log_2 y^4 + \log_2 y = 10$

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9 Solve for x in each of the following logarithmic equations:

a $\log_2 16 = x$

b $\log_{12} 144 = x$

c $\log_3 x = 4$

d $2 \log x = 4$

e $\log_{0.5} x = 10$

f $\log_{16} x = \frac{1}{4}$

g $\log_x 64 = 2$

h $\log_x 125 = 3$

i $\log_9 (7x + 5) = 4$

j $\log_6 (3x - 9) = 2$

k $\log_{10} (3x + 982) = 3$

l $6 \log_4 (2x) - 18 = 0$

m $11 \log_5 (x - 12) = 33$

n $\log_2 \left(\sqrt{2x^3}\right) + 1 = 4.5$

o $\log_2 (\log_2 x) = 0$

p $\log_2 (3x + 35) = \log_2 5$

q $\log_9 (10x - 3) = \log_9 (5x + 4)$

10 Solve for x in each of the following logarithmic equations:

a $\log_7 (9x - 2) = \log_7 (8x + 9)$

b $\log_6 x = \sqrt{\log_6 x}$

c $\log_2 x - 4 = \sqrt{\log_2 x - 4}$

d $3 \log x = \log 125$

e $\log_{10} x + \log_{10} 6 = \log_{10} 48$

f $\log_7 2x + \log_7 3 = 3$

g $\log_4 45 - \log_4 9x = 2$

h $\log_4 x = \log_4 7 - \log_4 (x - 4)$

i $\log x + \log (x + 3) = 1$

j $\log_3 x + \log_3 25x = 8$

k $\log_5 \left(4x^2\right) - \log_5 x = 3$

l $\log (x + 12) - \log (x + 5) = \log x$

m $\log_8 (5x + 12) = \log_8 (x + 6) - \log_8 3$

n $\log (x + 5) + \log (x - 2) = \log 8$

o $\log_4 (x + 3) + \log_4 (x - 3) = 2$

11 Solve for y in each of the following equations:

a $\ln (\ln e^{-y}) = \ln 3$

b $e^{-\ln y} = 3$

12 Solve the following equations, rounding your answers to two decimal places.

a $(4.5)^{\ln x} = 10^{0.5}$

b $\ln x = -\sqrt[3]{x + 8}$

13 Use a graphing calculator to solve the following equations. Round your answers to the two decimal places where necessary.

a $(\log_3 x)^2 - \log_3 \left(x^4\right) + 4 = 0$

b $3 \log^2 x - 2 \log x = 8$

c $\ln x = -\sqrt[3]{x + 9}$

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